A1. (45 minutes) (*15 points*) Consider a synchronous circuit that has a data input DIN, a clock input Clock, a reset input Reset, and a single output Z.

The output Z should be asserted true (set to 1) whenever the string of 1s and 0s entered since the last assertion of Reset, when interpreted as an unsigned binary integer, is divisible by 5. Otherwise it should be set to false (0).

Approach this problem using a Mealy machine. You can use the following properties:
1. If a%5=α and b%5=β then (a+b)%5=(α+β)%5.
2. If a current string has a numerical value of x then placing a 0 on the right hand side results in a string having a value x + x and placing a 1 results in a string having a value x+x+1.

Derive a symbolic state transition table for this problem (a Finite State Machine FSM). Choose a state encoding and then, using D flip-flops that are rising-edge triggered, derive expressions for the output Z and the new states as functions of the current states and the current inputs.

Give a:
a) Structural VHDL implementation of this FSM
b) Behavioral VHDL implementation of this FSM

B1. (45 minutes) (*15 points*) For the asynchronous Automaton described by the following equations:

$$Y_1 = y_1 y_0 x_1 + y_1 \overline{y_0} x_0 + \overline{y_1} y_0 \overline{x_1} x_0$$
$$Y_0 = x_1 x_0 + x_1 y_0$$
$$z = x_1 y_1 + x_0 \overline{y_0}$$

a) Draw the Automaton's schematic
b) Is this a Moore or a Mealy machine? Justify your answer.
c) Implement this automaton in VHDL
d) Determine its state transition table and outputs table
e) Draw its transition graph
f) Identify the potential problems in its functioning
g) Solve the potential problems in its functioning
h) Determine its new state transition table
i) Draw the Automaton's schematic

C1. (25 minutes) (*10 points*)

Examine if the following Automata are Lossless machines or not. In both cases, only if the Automaton is a Lossless machine, determine the input sequence knowing that the initial state, final state and output sequence were the ones given in the table:

a) Initial state = S_2

Final state = S_3

Output sequence = 110101

X States	0	1
S1	S2 / 0	S3 / 0
S2	S1 / 0	S4 / 1
S3	S5 / 0	S6 / 1
S4	S2 / 1	S3 / 1
S5	S6 / 1	S4 / 1
S6	S4 / 0	S6 / 0

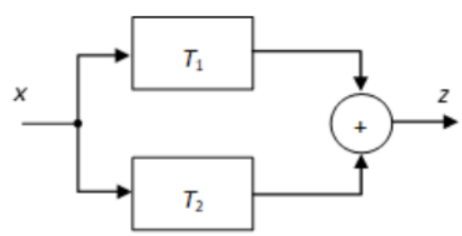
b) Initial state = S_5

Final state = S_1

Output sequence = 100101

X States	0	1
S1	S2 / 0	S3 / 0
S2	S1 / 0	S4 / 1
S3	S5 / 0	S6 / 1
S4	S2 / 1	S2 / 1
S5	S6 / 1	S4 / 1
S6	S4 / 0	S6 / 0

Be given the linear automata described by the following transfer functions, on GF(5):
$T_1 = 1 + D^3$; $T_2 = 1 + 2D + 3D^4$. The linear automaton T_3 is obtained according to the following schematic:



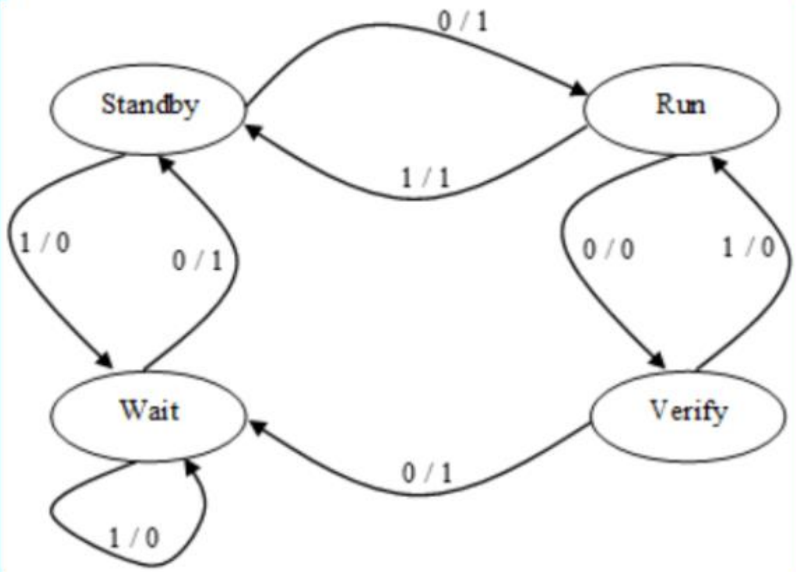
a) Obtain the transfer function of the new automaton T_3 and draw its block schematic;
b) Implement T3 using elementary components (logic gates, Flip-Flops etc.).
c) What is the impulse response of T_3?
d) Assuming T_3 is inertial, how will T_3 respond to the input sequence 3102?
e) Derive T_3's reverse automaton, T_3^{-1}, and draw its block schematic;

E1. (45 minutes) (*10 points*)

a) For the operations table of the microprogrammed Command Unit below, find the minimal set of command variables that must be sent to the Execution Unit (**D_0 – D_{13}** are the control signals of the different resources inside the Execution Unit):

Op	**D_0**	**D_1**	**D_2**	**D_3**	**D_4**	**D_5**	**D_6**	**D_7**	**D_8**	**D_9**	**D_{10}**	**D_{11}**	**D_{12}**	**D_{13}**
OP_0	0	X	0	0	0	X	0	0	0	0	0	0	0	0
OP_1	X	0	0	0	X	0	0	0	1	0	0	0	0	X
OP_2	0	0	1	0	0	0	1	1	0	1	1	X	X	X
OP_3	X	1	0	X	X	1	X	0	0	0	0	0	0	X
OP_4	X	0	X	0	1	0	1	0	0	0	0	0	0	X
OP_5	1	0	1	0	X	0	1	0	0	0	0	0	0	0
OP_6	1	0	X	0	X	0	0	0	0	0	0	0	0	1
OP_7	1	0	X	0	X	0	0	0	0	0	0	0	1	0

b) Consider the following automaton given by its transition graph:



a) Is this a Moore or a Mealy machine? Explain your answer.
b) Is this a connected automaton? Explain your answer.
c) Identify the automaton.
d) Implement this synchronous automaton in VHDL.
e) How would you implement it using Memories?